

A Highly Sensitive Tactile Sensor Using a Pyramid-Plug Structure for Detecting Pressure, Shear Force, and Torsion

Daehwan Choi, Sukjin Jang, Joo Sung Kim, Hyung-Jun Kim, Do Hwan Kim, and Jang-Yeon Kwon*

Sensors that detect and discriminate external mechanical forces are a principal component in the development of electronic tactile systems that can mimic the multifunctional properties of human skin. This study demonstrates a pyramid-plug structure for highly sensitive tactile sensors that enables them to detect pressure, shear force, and torsion. The device is composed of pyramid-patterned ionic gel inspired by neural mechanoreceptors and engraved electrodes. Based on a pyramid-plug structure, the deformation mechanism differs between different types of external mechanical loadings. The sensor provides the high sensitivities of 1.93 kPa^{-1} , 29.88 N^{-1} , and $3.39 (\text{N cm})^{-1}$, and a wide range of detection for tactile daily activity. Moreover, this tactile sensor could work through either of the two transduction methods (capacitive transduction and piezoresistive transduction). It is shown that this tactile sensor can be used to monitor changes in electrical signals ranging from those caused by human breathing to those caused by arbitrary multiplex human touching.

1. Introduction

An electronic sensory system that mimics the sensing capabilities of human skin is an attractive target for use in a broad range of applications, such as electronic skins, biomedical prosthetics, human-machine interface, remote surgery, and robotics. The ideal electronic sensory system model is a biological skin aided by a variety of neural mechanoreceptors for ionic mechanotransduction that sense various external stimuli such as pressure, shear force, torsion, vibration, and temperature.^[1–4] Of the several functions of these mechanoreceptors, their main role is detecting a broad range of tactile stimuli from the outside.^[5] This multifunctional tactile sensing ability is an essential factor for technology involving physical contact. For example, in robotics or prosthetics, it is important to recognize the magnitudes of the mechanical forces. While making contact with an object, we can obtain required information regarding grasp control, object manipulation, and friction determination

through the discrimination of multiple mechanical stimuli.^[3] Therefore, the ideal artificial electronic skin can discriminate external loads from various types of input and can also detect intensity.

Furthermore, the electronic skin must have properties that satisfy corresponding requirements for tactile sensing. For example, there are several relevant parameters that electronic skin sensors must measure, such as sensitivity, working range, detection time, and hysteresis. The measurement of these is important in order to realize the tactile sensing of human activities or motions. To demonstrate electronic skin with the capabilities of human-skin-like tactile sensing, many researchers have studied novel materials with improved sensing performance, including carbon nanotubes,^[6–12]

flexible elastomers,^[3,4,6,13–17] conducting polymers,^[14,18,19] ionic gels,^[5,7,20–23] nanowires^[24] or materials with the structure of nanoneedles,^[25] hemispheres,^[3,5,26,27] and pyramids.^[10,14,28–31] However, most of the previous studies have focused on the capability to perceive only a single type of mechanical stimulus, rendering previous attempts at designing an electronic skin incapable of sensing multiple forms of mechanical loads.

Previous studies have reported sensors with different materials and structural approaches for the detection and discrimination of the perception of multiple mechanical stimuli. Yin et al. developed the fiber strain sensor for tensile, bending, and torsion-sensitive properties using graphene-based composite fiber.^[32] In the present study, a fiber strain sensor was fabricated in a compression spring structure through a facile solution process with high reproducibility. Pang et al. presented an interlocking-based strain-gauge sensor for detecting multiple mechanical forces (including normal, tangential, and torsional forces) using Pt-coated microhair arrays.^[33] This strain sensor allowed for a simple and robust sensing platform using the interlocking structure for large-area sensing. Despite the potential of devices such as these, the sensors presented in previous reports exhibited low sensitivity for the perception of external mechanical stimuli in a low sensing range with only small electrical signal differences. They could also only detect a limited range of mechanical loadings.

To improve the performance of tactile sensors, a new tactile sensor should be designed by taking different approaches than the traditional approaches. In this study, we fabricated a tactile sensor composed of an ionic gel inspired by the ionic

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mechanical transduction phenomenon present in mammals, and an electrode using a geometric pyramid-plug structure for high performance and a high discriminating difference of external mechanical stimuli. Our tactile sensor could distinguish between and quantify multiple mechanical stimuli, measured for a wide range of normal pressures, shear forces, and torsional forces. We were also able to work the tactile sensor through each of the two transduction methods (capacitive and piezoresistive transduction) and studied the difference in the electrical signal caused by each transduction method.

2. Results and Discussion

2.1. The Pyramid-Plug Structure and the Deformation Mechanism of the Tactile Sensor

We used a pyramid-shaped ionic gel, which is an ionic and visco-poroelastic biocompatible elastomer, in order to mimic the phenomenon of ionic mechanotransduction found in the Merkel cells of mammals, which are abundant in the parts of the skin that are highly sensitive, such as in fingertips in humans. The ionic gel is an ionic thermoplastic polyurethane (i-TPU) composed of a polymer matrix (TPU) and ionic liquids (ILs), 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl) imide ([EMIM]⁺[TFSI]⁻, cation-anion pairs), which is a plasticizer in the TPU matrix that partially enables the intercalation of ions. The i-TPU layer was activated like the nanochannel of mechanoreceptors is in humans, inducing an ion-squeezing effect from the multiple mechanical loads, even at a low level of stimuli.^[5] This ionic gel was fabricated using a pyramid structure for the purpose of discriminating multiple mechanical loads through a simple fabrication process.

Figure 1a (top) shows a schematic illustration of the geometric design of which the ionic gel and the electrode were composed. The pyramid-shaped ionic gel film was placed

on an engraved iron electrode, resulting in what is called a pyramid-plug structure. The pyramid-engraved iron electrode [area ($5 \times 5 \text{ mm}^2$) and height ($\approx 3.5 \text{ mm}$)] was fabricated through mechanical processes. The pyramid-shaped ionic gel [area ($4.7 \times 4.7 \text{ mm}^2$) and height ($\approx 3.3 \text{ mm}$) of the pyramid] was prepared using an iron master molding process (see the Experimental Section). The active area of the tactile sensor was $1 \times 1 \text{ cm}^2$. Figure 1a (bottom) provides details about the inside of the ionic gel. The inside consisted of the polymer matrix and cation-anion pairs ([EMIM]⁺[TFSI]⁻).

As we applied the pyramid-plug structure to the tactile sensor, the mechanisms of the deformed pyramid-shaped ionic gel differed under external multiple mechanical stimuli as shown in Figure 1b–d. Figure 1b–d provides schematic illustrations of the deformation under different mechanical loads: (Figure 1b) normal pressure, (Figure 1c) shear force, and (Figure 1d) torsion. When normal pressure, such as a force moving up and down like in Figure 1b (right), is applied to the tactile sensor, the shape of the ionic gel becomes deformed, and the larger area of the gel in turn makes contact with the bottom electrode. Under shear force (also called a tangential force), such as a force pushing to one side like in Figure 1c (right), a shear stress is applied to the pyramid-shaped ionic gel. The ionic gel is moved to the side and deformed, and ends up making contact with the side area of the bottom electrode. Under a shear force larger than a shear friction, the ionic gel rises up diagonally. When the torsion, such as a rotational force like in Figure 1d (right), is applied to the sensor, the torque applied to the ionic gel film attempts to rotate the film. However, a friction, called a friction torque, is generated by the external torque at the contact area between the ionic gel and the bottom electrode. Under a torsional force larger than the friction torque, the ionic gel rises up horizontally.

Unlike the studies on the change in the electrical properties of ionic gel depending on the type and contents of the ionic liquids,^[34–37] we will study the change in the ionic gel brought on by external mechanical forces. Regarding the internal changes

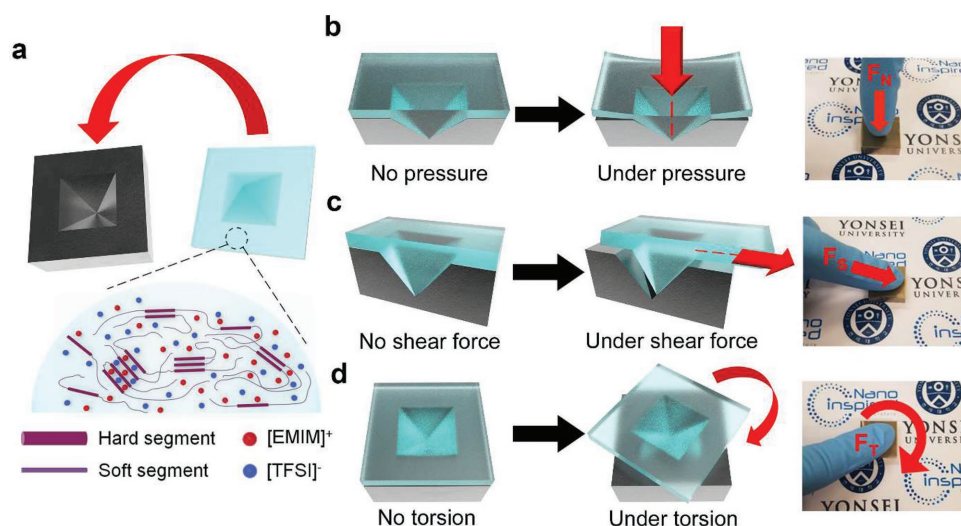


Figure 1. Tactile sensor based on geometric pyramid-plug structure of the ionic gel and iron electrode. a) (top) A schematic of the geometric structure of the electrode and ionic gel. (Bottom) Internal close-up of the ionic gel composed of the TPU matrix and ionic liquids. Schematic illustrations without stimulus and under three different mechanical stimuli for b) pressure, c) shear force, and d) torsion. There were possible geometric deformations of the pyramid-plug structure with mechanical loadings.

in the ionic gel, when the sensor was stimulated by external loads, ILs trapped inside the TPU matrix of the ionic gel came out through the soft segment of the polymer matrix and moved like free ions. This behavior is referred to as an ion-squeezing effect. Therefore, under external mechanical stimuli, ion squeezing was driven by the nanochannel of the TPU matrix.^[5] Free ions in the ionic gel then increased and an electric double layer (EDL) was established on the contact area between the ionic gel and the electrode. The electric signal under the external stimuli was larger than that without stimuli.^[5,7,21] By combining the pyramid-plug structure with a new material called ionic gel, the tactile sensor was able to detect and discriminate the type and magnitude of forces, because the mechanism of the deformation differed according to the type of the external mechanical stimuli as shown in Figure 1b–d.

2.2. Tactile Sensing Capability by Capacitive Transduction Method

To demonstrate different outputs from multiple mechanical forces, we measured the change in electrical properties as a

function of three different mechanical loadings: normal pressure, shear force, and torsional force. **Figure 2** provides experimental results from the tactile sensor of the pyramid-plug structure under different mechanical stimuli by the capacitive transduction method. The change in electrical signals in response to the normal pressure was measured with weights. An impedance analyzer (using the C_p mode) was wired to the ionic gel and the electrode. Figure 2a shows the capacitance change in response to normal pressure as a function of frequency from 10^6 to 20 Hz under AC bias 1 V. We recorded the change in capacitance from increasing the normal pressure (loading process) for 0–50 kPa. The variation in capacitance with the unloading process as a function of frequency is shown in Figure S3 (Supporting Information). At the lower frequency, the capacitance of the tactile sensor increased for all ranges when faced with the normal pressure. This frequency dependence of the ion-gel capacitance, where capacitance decreases with increasing frequency, is expected and known as capacitance dispersion. This behavior is caused by the interaction between electric charges on the solid electrode and counter-ions

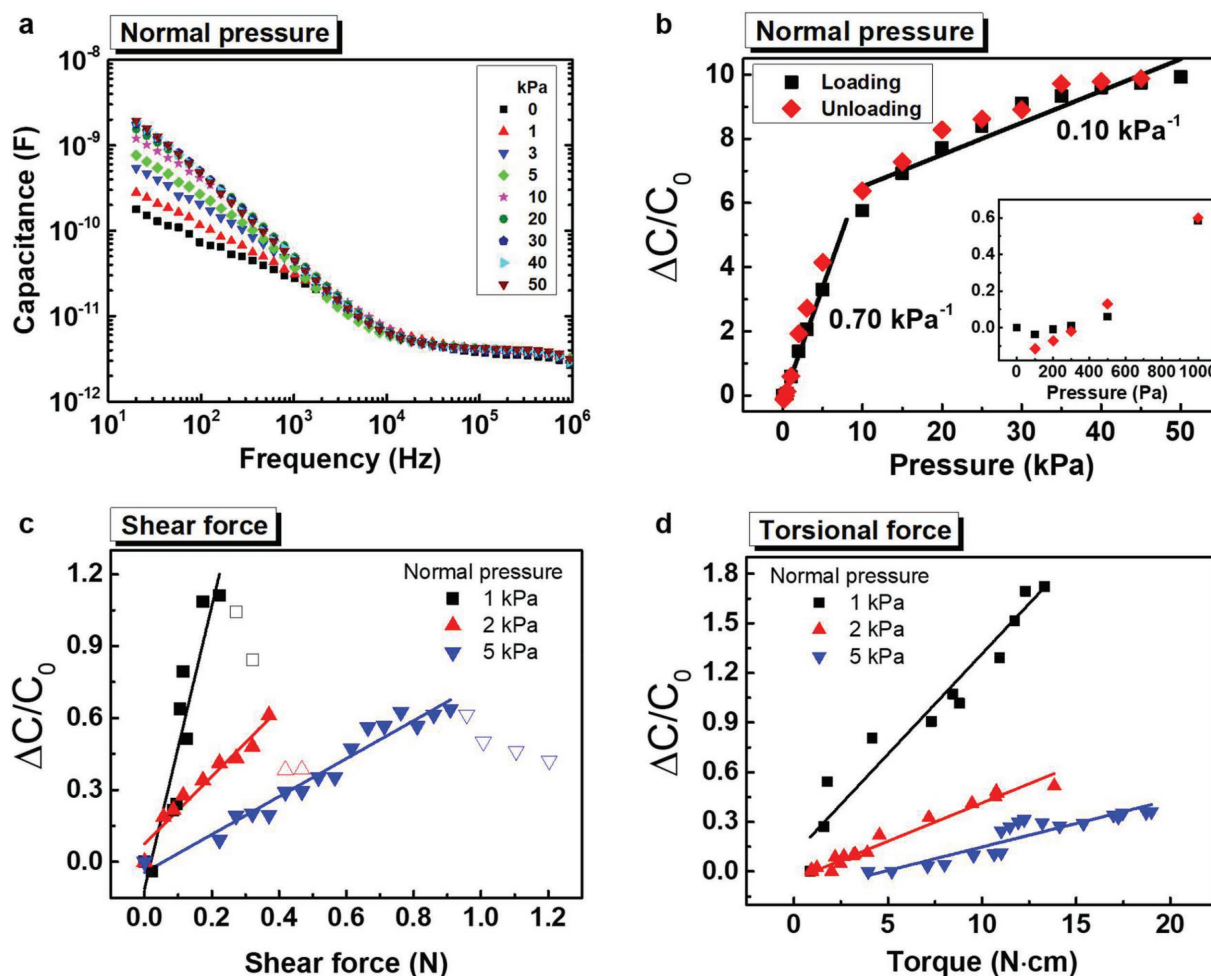


Figure 2. Sensing capabilities of the tactile sensor in response to normal pressure, shear force, and torsion loads by the capacitive transduction method. a) Capacitance in response to the normal pressure as a function of frequency at 20– 10^6 Hz. b) Pressure sensing capability of the tactile sensor with a pressure range of up to 50 kPa under AC bias 1 V at 20 Hz. The pressure sensitivity is 0.70 kPa^{-1} for 0–10 kPa. Inset: The detectable resolution is $\approx 100 \text{ Pa}$ with the loading–unloading process. c) Shear force sensing capability of the tactile sensor with a tangential force range of up to 1 N under AC bias 1 V at 20 Hz. d) Torsion sensing capability of the tactile sensor with a torsional force range of up to 20 N cm under AC bias 1 V at 20 Hz.

in the ionic gel, and is typical for ionic gels in contact with solid electrodes.^[21,34–36,38–40] As the normal pressure increases, the slope of the change in capacitance also increases, due to an increasing contact area caused by the deformation of the ionic gel and an increasing EDL establishment at the contact area.^[5] Thus, the difference in capacitance for the normal pressure at the low frequency was larger than at the high frequency due to the capacitance dispersion and the deformation.

Figure 2b shows the relative capacitance change of the pyramid-plug structure sensor as a function of the normal pressure at 20 Hz. The relative change in capacitance as a function of the normal pressure with different frequencies (20, 103, 498, and 1025 Hz) is provided as Figure S4 (Supporting Information). The slope of the plot refers to the pressure sensitivity (S_p), defined as $S_p = (\Delta C/C_0)/(\Delta P)$, where C and P are the capacitance and applied normal pressure, respectively. $\Delta C/C_0$ changed linearly under normal pressure of 0–10 kPa and 10–50 kPa, respectively. The pressure sensitivity of the pyramid-plug structure sensor with loading process was 0.70 kPa^{-1} for 0–10 kPa, which is a larger range than the range of pressure induced by gentle touch in daily activity (1–10 kPa). Regarding the unloading process, the relative change in capacitance had a similar trace to the loading process symmetrically. Measured from high pressure to low pressure, $\Delta C/C_0$ also changed linearly under normal pressure of 0–10 kPa and 10–50 kPa, respectively. The sensitivity for 0–10 kPa was 0.92 kPa^{-1} and that for 10–50 kPa was 0.10 kPa^{-1} , which is the same sensitivity of the loading process. Compared with other capacitive pressure sensors, the tactile sensor of the pyramid-plug structure had a comparable pressure sensitivity for a wide range of pressures (Table S1, Supporting Information). Figure S5 (Supporting Information) provides very little hysteresis of the loading and unloading process as a function of frequency with the normal pressures of 1 and 10 kPa. Provided in the inset of Figure 2b, the detectable resolution for normal pressure was 100 Pa, which is the same for pressure of 1 g on 1 cm^2 .

Figure 2c shows the relative capacitance change in response to shear force under AC bias 1 V at 20 Hz. We controlled the shear force that was applied to the tactile sensor by using a weight and recorded the performance of the device under the tangential force while applying different normal pressures of 1, 2, and 5 kPa (see the detailed measurement setup in Figure S1, Supporting Information). When the applied shear force increased, there was a linear increase of relative change in capacitance under different normal forces. For quantitative analysis, we define shear force sensitivity (S_s), $S_s = (\Delta C/C_0)/(\Delta F_s)$, where C and F_s are the capacitance and applied shear force, respectively.^[3] Under the normal pressure of 1 kPa, the shear sensitivity for up to 0.2 N was $\approx 5.89 \text{ N}^{-1}$, which is higher than that of other capacitive shear force sensors.^[18,33] Shown in Figure 2b, because the initial normal pressure affected the initial contact area and establishment of EDL, the sensitivity and sensing range could be manipulated by controlling the normal pressure to engage the pyramid-plug structure sensor.^[3] The smaller the initial normal pressure was, the smaller the initial capacitance (C_0) of the tactile sensor was; thus, according to the definition of shear force sensitivity, the shear sensitivity was much larger. And the stronger the pressing force was, the larger the induced frictional force was, allowing the tactile sensor

measure over a wider sensing range. In short, the lower the normal pressure was, the higher the shear force sensitivity was, but the sensing range also tended to decrease. The minimum detectable shear force was $\approx 0.05 \text{ N}$. This minimum detectable tangential force reflected the shear friction induced between the ionic gel and the electrode during the measurement for the shear force. The capacitance changes in response to frequency with different normal pressures (1 and 2 kPa) have a low hysteresis as shown in Figure S6 (Supporting Information).

Figure 2d plots the relative capacitance change in response to various normal pressures as a function of torque under AC bias 1 V at 20 Hz. We controlled the torque applied to the sensor using a force gauge and recorded the performance (see the detailed measurement setup in Figure S2, Supporting Information). As the applied torque increased, $\Delta C/C_0$ changed linearly for the entire range of measured torque to the tactile sensor. Since the initial electrical property of the tactile sensor was affected under the normal pressure, as in the measurement for the shear force, we measured with different normal pressures. We also define torsion sensitivity (S_T) for different normal pressure loadings as $S_T = (\Delta C/C_0)/(\Delta F_T)$, where C and F_T are the capacitance and applied torsional force, respectively. Similar to the electrical response under the shear force, the tactile sensor had a tendency to decrease the torsion sensitivity and increase the sensing range when the normal pressure increased. When the normal pressure was 1 kPa, the torsion sensitivity was $0.12 (\text{N cm})^{-1}$ for 0–12.3 N cm and the minimum detectable torque was $\approx 0.87 \text{ N cm}$ as shown in Figure S7 (Supporting Information). The minimum detectable torque changed with the variations in normal pressure. Because the friction torque that was generated in the contact area between the ionic gel and the electrode increased when the normal pressure increased, torsion sensing was possible with a larger torque.

We measured the capacitance change in response to different mechanical stimuli. It is noted that this pyramid-plug sensor could discriminate the type of external mechanical loading by the capacitive transduction method. In the capacitive method, our experimental results demonstrated that the sensitivities were 0.92 kPa^{-1} (normal pressure), 5.89 N^{-1} (shear force), and $0.12 (\text{N cm})^{-1}$ (torsion), performances of which are reasonably comparable or superior to previous capacitive tactile sensors.^[18] Notably, this tactile sensor can work through the capacitive method unlike previously reported tactile sensors, which are only able to discriminate mechanical stimuli based on the piezoresistive transduction method.^[32,33]

2.3. Tactile Sensing Capability by Piezoresistive Transduction Method

We also recorded the electrical signal of the pyramid-plug structure sensor through the piezoresistive transduction method. We expected that the electrical signal change or pattern brought on by the piezoresistive method would differ from that of the capacitive method. The setup for the measurement of each stimulus was the same as that of the capacitive transduction method. The variation in electrical current was recorded with different external mechanical loadings using a sourcemeter.

The relative current change in response to normal pressure under DC bias 1 V with increasing pressure is shown in

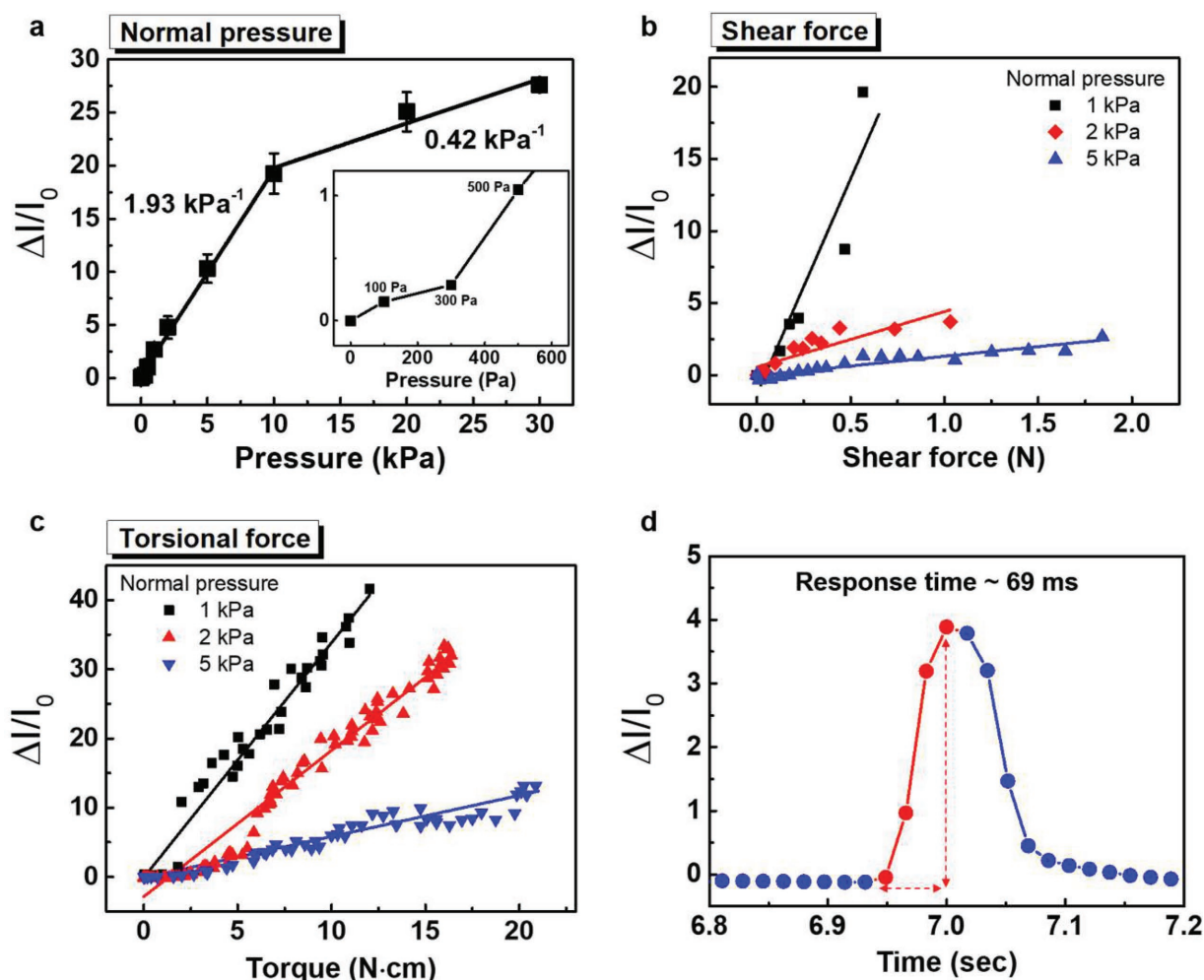


Figure 3. Sensing capabilities of the tactile sensor in response to normal pressure, shear force, and torsion loads by the piezoresistive transduction method. a) The current in response to normal pressure as a function of pressure with a pressure range of up to 30 kPa under DC bias 1 V. The pressure sensitivity is 1.93 kPa^{-1} for 0–10 kPa and 0.42 kPa^{-1} for 10–30 kPa. Inset: The detectable resolution with the loading process. b) Shear force sensing capability of the tactile sensor with a tangential force range of up to $\approx 2 \text{ N}$ under DC bias 1 V. c) Torsion sensing capability of the tactile sensor with a torsional force range of up to 20 N cm under DC bias 1 V. d) Real-time monitoring of the relative current change in response to an air blowing. The response time is $\approx 69 \text{ ms}$.

Figure 3a. We also measured the current change to normal pressure with the unloading process as shown in Figure S8 (Supporting Information). As the applied normal pressure increased to 30 kPa, $\Delta I/I_0$ also increased. The change in current of the tactile sensor increased linearly from 0 to 10 kPa, while the slope of the plot decreased at the pressure of more than 10 kPa. The slope of the plot was the pressure sensitivity, like in the capacitive transduction method. The normal pressure sensitivity of the tactile sensor was 1.93 kPa^{-1} for 0–10 kPa. For the pressure range of 10–30 kPa, a range used during object manipulation, the sensitivity was 0.42 kPa^{-1} (comparison to the sensitivity of other piezoresistive pressure sensors is provided in Table S2 of the Supporting Information). As shown in the inset graph of Figure 3a, the minimum detection resolution was $\approx 100 \text{ Pa}$, which is much smaller than that of the gentle touch of 1–10 kPa.

Figure 3b shows the relative change in current as a function of shear force with different normal pressures under DC bias

1 V. As the shear force increased, the relative current change linearly increased up to a certain level. The minimum detectable shear force was $\approx 0.026 \text{ N}$, which is half of that observed by the capacitive measurement method. This difference in the detection limit for the shear force depending on the transduction method used indicates that the piezoresistive transduction method responded faster to external shear force and could detect changes in the electrical signal at even a smaller stimulus than the capacitive method could. The largest shear force sensitivity was 29.88 N^{-1} for the normal pressure of 1 kPa, which is ≈ 22.1 times the sensitivity (1.35 N^{-1}) at the normal pressure of 5 kPa. This behavior can be explained by the decreased initial contact area and decreased free ions of the deformed ionic gel caused by a lower normal pressure, providing the possibility to further increase free ions of the ionic gel by applying a shear force.^[3,12]

Figure 3c provides the details of the torsion-sensing capability of the tactile sensor using the piezoresistive transduction

method for different normal pressures under DC bias 1 V. As the applied torque increased, the relative change in current of the tactile sensor linearly increased as well. The minimum detectable torque was ≈ 0.04 N cm. The largest torsion sensitivity was 3.39 (N cm) $^{-1}$ for the normal pressure of 1 kPa, which is ≈ 5.38 times the sensitivity (0.63 (N cm) $^{-1}$) at the normal pressure of 5 kPa. Like the electrical response to the capacitive transduction method, the sensitivity and sensing range could be manipulated by controlling the normal force to engage the pyramid-plug structure.^[3,12]

Figure 3d provides the response of the tactile sensor to arbitrary air blowing as a function of time. By simply blowing air, the relative current change rapidly increased to the peak and then decreased to the initial state. The response time of the tactile sensor was ≈ 69 ms, which is comparable to other piezoresistive pressure sensors (Table S2, Supporting Information).

We also studied the performance of the tactile sensor in the piezoresistive transduction method according to different mechanical loadings with higher sensitivity than other multifunctional tactile sensors. More clearly, although the sensitivity of our tactile sensor to normal pressure is not very high as compared with other pressure sensors in Tables S1 and S2

(Supporting Information), our sensor has a much higher sensitivity to shear force and torsional force than other tactile sensors responsive to multi mechanical stimuli.^[3,12,32,33] It should be noted that the discrimination by type of mechanical stimuli performed by our tactile sensor could work through each of the two transduction methods (capacitive and piezoresistive) in a wide range of pressures, shear forces, and torsions with a single-device platform.

2.4. Differences in Electrical Properties by Each of the Transduction Methods

To explain the difference observed by the type of electrical signal, Figure 4 provides characterizations of electrical properties by two different transduction methods (capacitive and piezoresistive) with various mechanical stimuli. Figure 4a shows the relative change in electrical properties as a function of normal pressure for 0–30 kPa. In both methods, the value of the relative change increased from 0 to 10 kPa with increases in the normal pressure, and the slopes of the plots decreased from a high of 10 kPa. This tendency of the slope

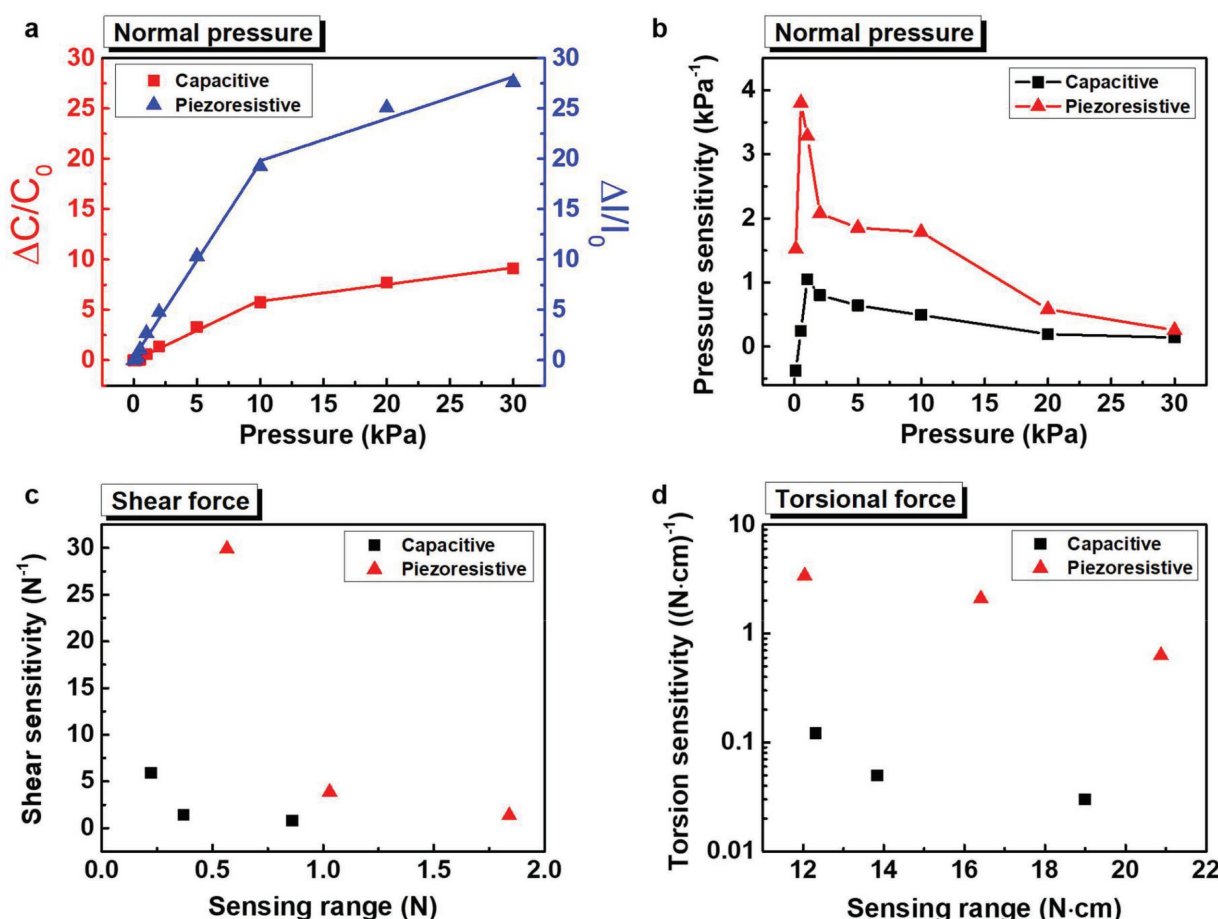


Figure 4. Characterizations of electrical responses to multiple mechanical loadings by two transduction methods (capacitive and piezoresistive transduction methods). a) Comparison of the two measurement methods in response to normal pressure for 0–30 kPa. b) Differential values of pressure sensitivity at each point. c) By the two measurement methods, the difference of the shear force sensitivity for the tangential force versus sensing range. d) Comparison of the torsion sensitivity as a function of sensing range by the two transduction methods.

decreasing at a certain point was very similar to that of the pressure sensors using pyramid-patterned materials.^[29,30] For a range of 0–10 kPa, the sensitivity of the piezoresistive method was 1.93 kPa^{-1} , which is ≈ 2.75 times the sensitivity (0.70 kPa^{-1}) elicited by the capacitive method. In the case of the capacitive method, the capacitance of the ionic gel at low frequency was affected by the establishment of EDL, which appears on the surface of the contact area as one to two layers inside the ionic gel. After the formation of one to two layers, as indicated by the Helmholtz model, the contact area saw no further development of the EDL because the electrical potential drops with increasing distance from the interface of electrode/ionic gel.^[41] Therefore, when the stimulus was above a certain level (in the case of the normal pressure, $>10 \text{ kPa}$), the capacitance did not increase by too much more, as shown in Figure 4a (the sensitivity is 0.17 kPa^{-1} for 10–30 kPa).

However, in the case of the piezoresistive method, since free ions squeezed out from the i-TPU matrix increased greatly with increasing pressure, the deformed ionic gel with an increase of the high-density free ions that are ionic liquids (not bounded to TPU chains, moving freely) led to an increasing current flow.^[5,22] Therefore, the relative electrical change of the piezoresistive method was much higher than that of the capacitive method. The relative change in electrical current measured as a function of normal pressure in the case of using pristine TPU and the ionic gel is provided as Figure S8 (Supporting Information). In the case of a pristine TPU, as the normal pressure increased, electrical signals increased only by increasing contact area, not free ions, because it did not contain ionic liquids. It can be seen that the relative change of the ionic gel in which free ions were present by the ion-squeezing effect was higher than that of the pristine TPU. Therefore, the amounts of free ions induced by external stimuli inside the ionic gel would be the main factor resulting in this higher sensitivity than the contact area between the ionic gel and the electrode.^[22]

For an explanation of the different electrical signal intensity induced by the two transduction methods, Figure 4b provides the details of the change in instantaneous rate of the slope in response to the normal pressure. The instantaneous sensitivity of the piezoresistive method was maintained for a range of 0–20 kPa at a higher level than that of the capacitive method. Meanwhile, when the external loading of $\approx 20 \text{ kPa}$ or more was applied to the tactile sensor, the rate of the change in the instantaneous sensitivity by the two transduction methods would be at a similar level. Also, the point where the instantaneous sensitivity increased rapidly differed between the two transduction methods. The peak of the instantaneous sensitivity with the piezoresistive method was at a smaller stimulus (3.80 kPa^{-1} at 500 Pa) than that with the capacitive method (1.05 kPa^{-1} at 1000 Pa). This phenomenon indicates that the piezoresistive method responded to the external mechanical loading faster than the capacitive method. In other words, the piezoresistive method had the possibility to lead to an advantage in sensitivity and minimum detectable stimulus compared to the capacitive method.

Figure 4c plots the difference of the sensitivity versus the sensing range for shear force by the two transduction methods. We observed that the shear force sensitivity increased with a decrease in the normal pressure as shown in Figures 2c and 3b.

The largest shear force sensitivity found was 29.88 N^{-1} by the piezoresistive method for the normal pressure of 1 kPa , which was ≈ 5.07 times the sensitivity (5.89 N^{-1}) of the same by the capacitive method. With the same normal pressure, the maximum detectable shear force by the piezoresistive method was much larger than that by the capacitive method.

Figure 4d shows a comparison of the sensitivity for the torsional force as a function of the sensing range by the different transduction methods. Similar to the case of the electrical response to the shear force, we also observed that the torsion sensitivity increased with a decrease in the normal pressure, as shown in Figures 2d and 3c. In the case of torsion sensing, the difference in sensitivity between the two transduction methods was much larger than that in the other mechanical stimuli. The largest torsion sensitivity was $3.39 (\text{N cm})^{-1}$ by the piezoresistive method for the normal pressure of 1 kPa , which was ≈ 28 times the sensitivity ($0.12 (\text{N cm})^{-1}$) of that by the capacitive method. The torsion sensitivities were much higher than that of the other tactile sensors.^[33]

For normal pressure, the contact area between the electrode and the ionic gel increased as the pressure increased, and the ion-squeezing effect also increased. However, by the pyramid-plug structure of the tactile sensor, in the case of the shear force, when the external stimulus increased, the ionic gel moved to one side and rose up diagonally as shown in Figure 1c. In the case of the torsional force, the ionic gel rose up horizontally as shown in Figure 1d. In other words, the contact area decreased as the stronger shear force or torsional force was. Nevertheless, the relative capacitance and current in response to external mechanical stimuli increased as shown in Figures 2c,d and 3b,c. It indicates that the effect of free ions was the main factor to the change in the electrical properties than the variations of the contact area. The reason for the higher sensitivity of the piezoresistive method was that the increased amounts of free ions induced by the ion-squeezing effect would effectively affect electrical current; therefore, it resulted in the higher sensitivity than that of the capacitive method.

2.5. Applications of the Pyramid-Plug Tactile Sensor

As a further example, the periodic respiration of a volunteer was monitored in real time. As shown in Figure 5a, periodic breathing can generate an obvious periodic change in the electrical current of the tactile sensor. Additionally, breathing mode—normal ($\approx 200 \text{ Pa}$) and deep ($\approx 500 \text{ Pa}$)—can be easily distinguished from the relative current change, suggesting potential applications of the tactile sensor in monitoring different breath patterns depending on the person being monitored.^[11]

Since the tactile sensor of the pyramid-plug geometry showed different patterns of deformation and electrical intensities specific to the types of external mechanical forces, the tactile sensor would be able to detect and discriminate the type of mechanical stimuli. To demonstrate its capability in measuring multiplex mechanical loads, the electrical signal for the finger touch as a signal of each of the three different physical inputs was monitored in real time.

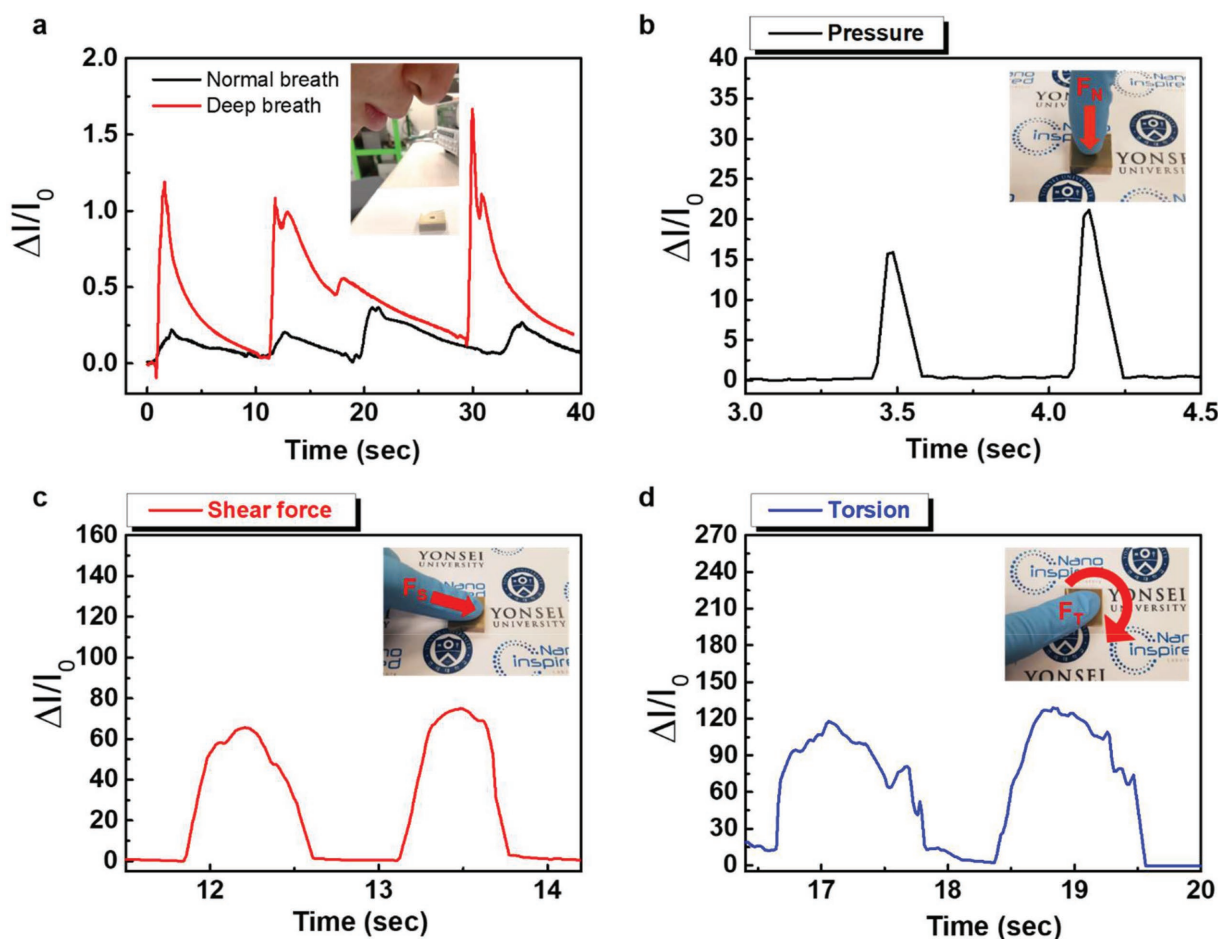


Figure 5. Real-time measurement in applications of the tactile sensor for the detection and differentiation of multiple mechanical stimuli. a) The relative current change of the tactile sensor in response to human breathing mode. Real-time monitoring of the relative current change by a finger touch in response to dynamic loading and unloading cycles of b) normal pressure, c) shear force, and d) torsion.

Figure 5b–d provides the details of the different patterns of deformation depending on (b) normal pressure (black), (c) shear force (red), and (d) torsion (blue). The tactile sensor showed different electrical signal patterns and intensities for the different types of mechanical stimuli. As can be seen in Figure 5b–d, the reason why the electrical signals for different types of mechanical stimulus have a different level is that the shear force ($\Delta I/I_0 < 80$) and torsion ($\Delta I/I_0 < 180$) were mechanical loadings that are affected by the normal pressure ($\Delta I/I_0 < 40$). If I_0 , the initial current, is set to the same value as that of the current without pressure, the relative current change for the tangential force or torsional force is much higher than that for the normal pressure. Thus, it was noted that the pattern of the variation in the current depending on different external mechanical forces could be differentiated by different signal intensities and patterns.

3. Conclusions

A highly sensitive tactile sensor based on a pyramid-plug structure allows us to efficiently discriminate a wide range of mechanical stimuli in a single-device platform. Here, each of the relative electrical changes under different mechanical

stimuli showed a unique output pattern to the specific type of mechanical forces, leading to the possibility of discriminating multiple mechanical stimuli. Notably, we have demonstrated a tactile sensor, which works by each of the two electrical transduction modes, with sensing capabilities of three mechanical loadings (pressure, shear force, and torsion), and possessing the combination of high sensitivity and a broad sensing range. Moreover, this tactile sensor was fabricated by a facile, simple, and scalable process that is applicable for use in the practical and widespread wearable sensors. In contrast to other sensing platforms, the pyramid-plug tactile sensor requires no complex integrated system or layered arrays. The combination of the ionic gel mimicking the ionic mechanotransduction mechanism in human skin and the structure of the pyramid-plug geometry brings about high sensitivity over a wide range of various tactile activities from gentle touch to object manipulation, which indicates that the tactile sensor may possibly replace the role of various mechanoreceptors as an ideal sensory system for full-range tactile sensing. This tactile sensor with a pyramid-plug geometry has the potential to be used as a complex-utility electronic skin for robotics, health-monitoring devices, and prosthetics, and even as an implantable device if biocompatible electrical materials could be integrated.

4. Experimental Section

Fabrication of Pyramid-Shaped Iron Electrode and Ionic Gel: The fabrication of the pyramid-plug structured iron electrode and ionic gel involved (i) fabrication of the iron master and (ii) preparation of the ionic gel solution.

- (i) **Fabrication of the iron master:** The engraved iron electrode of the pyramid structure was fabricated through discharging, etching, and mechanical milling. The dimensions of the pyramid design of the iron master were an area of $5 \times 5 \text{ mm}^2$ and a height of 3.5 mm.
- (ii) **Preparation of ionic gel solution:** TPU chips (KA-480) were obtained from Kolon Industries, Inc. Ionic liquid ([EMIM]⁺[TFSI]⁻) and *N,N*-dimethylformamide (DMF) were supplied by Sigma-Aldrich. First, the TPU chips and DMF were mixed in a 4:1 ratio (w/w) and stirred at 120 °C for 2 h. Then, 20 wt% [EMIM]⁺[TFSI]⁻ based on the weight of the TPU chips was added into the solution and stirred for 30 min at 150 °C.
- (iii) **Fabrication of the pyramid-patterned ionic gel:** The ionic gel films were fabricated through a drop-casting method. The blended solution was degassed and poured on the iron master and hardened in an oven at 120 °C. During the curing process, the ionic gel films were placed in an oven for 24 h in order to fully remove the residual solvent. The size of the ionic gel prepared using the iron mold was $4.7 \times 4.7 \text{ mm}^2$ with a 3.3 mm height and a 0.7 mm thickness. The cured pyramid-patterned ionic gel was peeled off from the master, and then the iron master served as the bottom electrode of the tactile sensor.

Measurement for the Electrical Properties: The capacitive properties of the tactile sensor were measured using a two-probe method (Impedance analyzer E4990 A, Keysight Technologies) at AC 1 V. The piezoresistive properties of the tactile sensor were measured using a sourcemeter (KEITHLEY 2601 A) at DC 1 V. In order to apply mechanical forces on the tactile sensor, weights were used to provide normal forces and shear forces. A force gauge (Mark-10, M5-05) connected to a computer was used to apply torque to the sensor on a custom-built probe station (using probe station 8000, MS TECH).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

deformable ionic gel, electronic skin, mechanotransducer, micro-structured, tactile sensor

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